# Time reversal is the one twist a self-referential code cannot enact: a logical-holonomy census of the [[5, 1, 3]] code

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**Abstract**

An embedded observer who can only read and re-encode a quantum system — never step outside it — can enact only unitary operations on its own logical data. We make this concrete for the $[[5, 1, 3]]$ perfect code. A closed loop in code-deformation space is an automorphism of the stabilizer group realized by a qubit permutation composed with a transversal single-qubit Clifford; its induced action on the logical Bloch frame is a *holonomy*. Enumerating all $120 \times 24 = 2880$ candidate loops, the 240 automorphisms realize exactly 24 distinct logical holonomies. They form the octahedral group $S_4$ — the full logical Clifford group modulo Pauli's — and *every one is a proper rotation*, $\det R = +1$. The single transformation absent from this group, orientation reversal of the Bloch frame, is *antiunitary*; by Wigner's theorem the antiunitary symmetry of quantum mechanics is time reversal. Hence the one twist an embedded self-reader cannot acquire by any closed self-transport is running the film backwards. We verify the nontrivial (unitary) holonomy on superconducting hardware: a 120° logical-frame rotation is measured with $+0.898$ overlap against the rotated prediction versus $-0.003$ unrotated (IBM `ibm_kingston`, job `d958plnu62ks7396rgmg`). The exclusion itself is generic to any unitary framework; our contribution is the concrete, hardware-backed instantiation that the *reachable* inside-twists form a definite nontrivial group whose single missing element is exactly time reversal.

## 1 Setup: what twists can an inside observer acquire?

This note isolates one sharp consequence of a recurring theme in the Executable Positive Geometry program: every structure the program builds is timeless, spatial relations are freely tradeable, and temporal orientation is the quantity that resists being manufactured or reversed from within a self-contained description. Here "from within" is made precise as the group of operations available to an observer who is a subsystem of a quantum error-correcting code and can only permute and locally re-encode the physical qubits while keeping the code — the rulebook of self-description — invariant.

The $[[5, 1, 3]]$ code encodes one logical qubit in five physical qubits and is the smallest perfect code [1]; it is also the elementary tile of the HaPPY holographic code [2], so its logical data is a toy bulk degree of freedom reconstructed from a boundary. A *code-deformation loop* is an operation

$$W \;=\; P_\sigma \bigotimes_{j=1}^{5} U_j, \qquad U_j \in \mathrm{Clifford}_1, \quad \sigma \in S_5, \tag{1}$$

a physical-qubit permutation $P_\sigma$ composed with a transversal single-qubit Clifford, chosen so that $W$ maps the stabilizer group of the code *exactly back to itself*. The code returns home; the logical qubit need not. The induced logical operation $R(W) \in \mathrm{SO}(3)$ or $\mathrm{O}(3)$ acting on the Bloch frame is the loop's **holonomy**: the measurable residue of transporting self-knowledge around a closed path of self-descriptions. The Klein-bottle question that motivated the census is sharp: does any closed loop *reverse* the orientation of the logical frame, $\det R = -1$?

## 2 The census

We enumerate all $|S_5| \times |\mathrm{Clifford}_1| = 120 \times 24 = 2880$ candidate loops, retain those that are genuine stabilizer automorphisms, compute each one's action on the logical frame, and assemble the holonomy group. The computation is exact (statevector arithmetic on the codewords built in `happy_tile.py`); it is reproduced by `holonomy.py` in the repository.

**Result.** Of the 2880 candidates, 240 are automorphisms. Their logical actions collapse to 24 **distinct holonomies**, with element orders

$$\{\,1{:}1,\ 2{:}9,\ 3{:}8,\ 4{:}6\,\}, \tag{2}$$

the conjugacy-class profile of the octahedral group $S_4$. This is precisely the logical Clifford group modulo Pauli's, $\mathrm{Clifford}_1/\mathrm{Pauli} \cong S_4$: closed journeys through rulebook-space realize the full single-qubit Clifford symmetry of the logical frame. Sample elements include 120° rotations (order 3), 180° rotations (order 2), and 90° rotations (order 4).

**The determinant is always +1.** Every one of the 24 holonomies is a *proper* rotation:

$$\det R(W) = +1 \quad \text{for all automorphisms } W. \tag{3}$$

No closed loop in code space reverses the orientation of the logical Bloch frame. The reachable inside-twists form a genuine nontrivial rotation group, but that group is SO(3)-valued: it never touches the orientation-reversing component of O(3).

## 3 The missing twist is time reversal

The absence of $\det R = -1$ is not an accident of this code, and we are careful to say so. Unitary conjugation $\rho \mapsto U\rho U^\dagger$ acts on the Bloch sphere by proper rotations only; orientation reversal is implemented by an *antiunitary* operator. By Wigner's theorem [3], every symmetry of quantum mechanics is either unitary or antiunitary, and the canonical antiunitary symmetry — the one whose square structure and complex conjugation single it out — is time reversal $T$. An observer confined to unitary self-operations therefore cannot, by *any* closed self-transport, acquire the mirror twist:

$$\boxed{\text{reachable inside-holonomies} = S_4 \subset \mathrm{SO}(3); \qquad \text{missing element} = \text{orientation reversal} = T.} \tag{4}$$

The one operation reserved for a vantage outside the unitary description is the one that runs the film backwards. In the program's slogan, temporal orientation is the currency of self-containment: the inside-operations form a group with no orientation-reversing element, and the single element they cannot reach is precisely the arrow's reversal.

## 4 Hardware witness

The *nontrivial* content — that the reachable holonomies form a real rotation group rather than collapsing to the identity — is not guaranteed by unitarity and is worth measuring. We prepared the $[[5, 1, 3]]$ tile on `ibm_kingston` and applied a transversal Clifford layer realizing an order-3 automorphism (42 two-qubit gates after heavy-hex routing). The logical frame rotates by 120°: the post-loop logical state overlaps the *rotated* prediction with $+0.898$ and the *unrotated* reference with $-0.003$ (IBM job `d958plnu62ks7396rgmg`). The twist is real and it is a proper rotation; the mirror twist is, as predicted, nowhere on the device.

## 5 Relation to prior work, and honest scope

The statement "unitary conjugation cannot implement time reversal" is textbook. What is, to our knowledge, unstaked is the concrete finite-code, hardware-backed form of it: that for an explicit self-referential code the reachable inside-holonomies are exactly $S_4$ and the *single* missing element is $T$. Three operational results frame the same boundary from different angles and should be read alongside this note. Manikandan and Jordan [4] characterize the time-reversal symmetry of generalized measurements with past and future boundary conditions. Di Biagio, Donà and Rovelli [5] locate the operational arrow of time in assumptions about which quantities are known to an agent. Chiribella, Aurell and Życzkowski [6] prove a no-go for extending the time reversal of unitary dynamics to all quantum evolutions, with conditioning on past outcomes recovering the standard operational theory. Our result is complementary and concrete: a physical code, an exhaustive holonomy census, and a measured twist.

**Scope.** We state the limitation plainly. The *exclusion* of the mirror twist is generic to any unitary framework — it is Wigner's theorem, not a property of the $[[5, 1, 3]]$ code — so this note does not claim a new theorem about time. Its contribution is threefold and entirely concrete: (i) the reachable inside-twists are computed exactly and shown to form the specific nontrivial group $S_4$, the full logical Clifford symmetry; (ii) the single unreachable element is identified as time reversal, tying the census to the program's temporal- orientation hypothesis; and (iii) a nontrivial element is realized and measured on hardware. The result is a witness, made physical, of where the boundary of self-enactable operations lies — and that the boundary is drawn exactly at the arrow of time.

## References


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